

Is it for Real?

Yes, all of these bandwidth figures and data pumping numbers are only theoretical. In real life, the performance increase that you should expect from DDR-based machines would be generally in the range of 5 to 7 per cent at the most. Realistically speaking, a non-discernible performance increase is what you should expect in ordinary desktop mode. Nevertheless, a substantial performance increase should be expected in memory intensive applications such as Photoshop and CAD-based applications.

So does it make sense to go out and buy some DDR and put it in your system? No.

DDR is not backward compatible with SDRAM slots. It uses a 184-pin interface and has only one notch in-between, whereas SDRAM has two notches and uses a 168-pin interface. But if you are buying a new system then DDR-SDRAM is definitely worth considering (or waiting for, if you want an Intel-based machine).

Also consider that there could be incompatibility issues when different modules from different manufacturers are used. Though these issues have been reduced, they may continue to exist because of certain memory verification issues. Always use both modules from the same manufacturer. Remember that the price difference between ordinary SDRAM and DDR isn't too much and this applies to PC2100 (PC266, which is the fastest available).

around to mustering enough support—its performance did not show any substantial increase and the high pricing made it unattractive to most people. DDR-SDRAM, on the other hand, got a lot of manufacturers and vendors excited because the concept by itself was pretty straightforward and the prices touted were in line with the prevailing prices of SDRAM. Simply put, it was almost what everyone wanted SDRAM to become—faster and cheaper.

What makes it tick?

DDR-SDRAM, as the name suggests, is a variant of SDRAM. Fundamentally, DDR-RAM is like any normal SDRAM that it sends twice the amount of data in one clock cycle. This is accomplished by sending data on the rising edge of the clock cycle as well as on the falling edge. But it is not as simple as it sounds. Pushing a normal SDRAM module to function at twice its capability involves a great deal more.

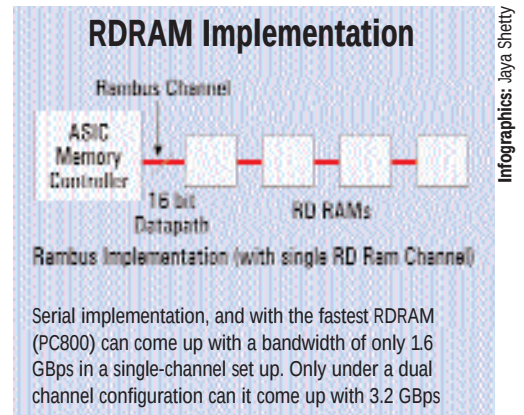
The term 'double data rate' refers to the fact that the theoretical peak data rate is twice the rate at which commands can be clocked. Specifically, commands are entered on the positive edges of clock, and data occurs for both positive and negative edges of clock. In most applications, the relative advantages of this approach are obvious. Doubling the transfer rate on the data bus and not changing the transfer rate on the command and address bus is a relatively simple approach to doubling memory bandwidth.

However, to understand the complete internal operations, it is important to think of DDR SDRAM as being more than just double data rate; the term DDR should be primarily associated with a 2n-prefetch, double data rate device. The terminology '2n-prefetch' implies that the internal data bus is twice the width of the external data bus and, therefore, the internal column access frequency can be half the external data

transfer rate. This means that for each single read access cycle internal to the device, there are two external data words provided.

What it means

Think of a memory chip as an Excel spreadsheet in which the different bits of information are stored temporarily in columns and rows. The access pattern or strobe follows this organisation in that first a field within the first column is accessed and subsequently all other addresses



within that row can be read without additional penalty cycles. Similarly, two external data words written to the device are internally combined and written in one internal access.

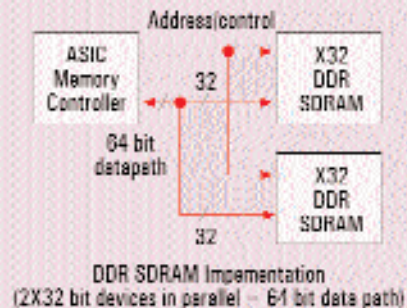
To the layman, this means that data access occurs in pairs, that is, a single read access fetches two data WORDs (of 32-bits each); and for a single write access, two data WORDs (and/or data mask bits) must be provided.

Only doubling data bandwidth while leaving the core column access frequency the same as SDR has important implications. One extreme is an SDR SDRAM (1n-prefetch) running at literally twice the current frequency. But this is physically restrictive since it is much more difficult for the system designer to execute the command/address bus, and it is more costly for the DRAM vendor because it requires a larger die size and a more expensive process in order to double the internal core frequency of the DRAM. At the other extreme of possibilities is an 8n-prefetch architecture that provides increased bandwidth even with a narrow external memory data bus. However, defining a narrow external bus at the expense of a very wide internal bus increases the cost of the SDRAM device due to additional die size and process steps. Between these two extremes lies DDR, which is probably the best compromise considering technical and cost reasons.

Doing it two-gether

Of course, all these subtle nuances of prefetch will be of no use if it does not translate into

DDR-SDRAM Implementation



Parallel implementation under single channel combination comes up with a maximum bandwidth of 2.1 GBps, which would double in a dual channel setup